

Introduction to Machine Learning Methods

Supervised learning

Haris Gačanin

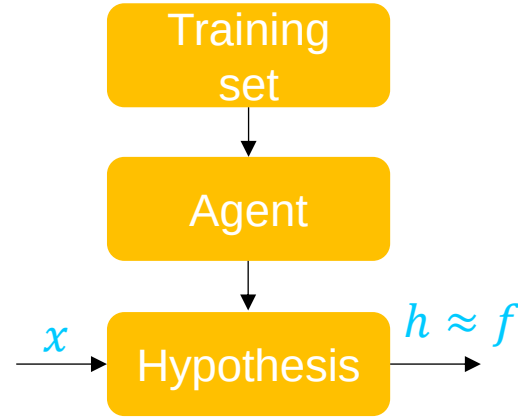
RWTH Aachen University

Outline

- [illegible]

Supervised Learning Principle

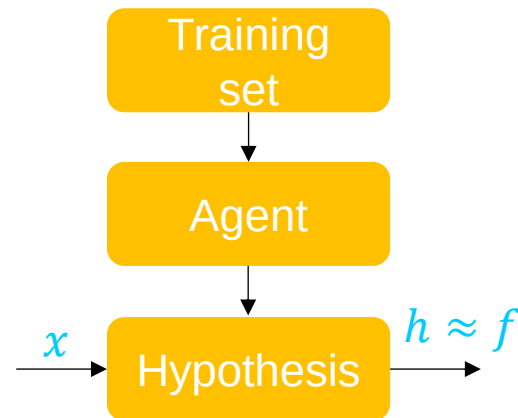
- Task of supervised learning is:
 - Given a training set of n examples $\{x^{(i)}, y^{(i)} = f(x^{(i)})\}$ where $y^{(i)}$ was generated by unknown function f
 - discover a function h (hypothesis) that approximates the true function f (i.e. $h \approx f$)
- h maps x 's to y 's



Learning is a search through the space of possible hypotheses $\mathcal{H}$ for one that will perform well, even on new examples beyond the training set

Remarks

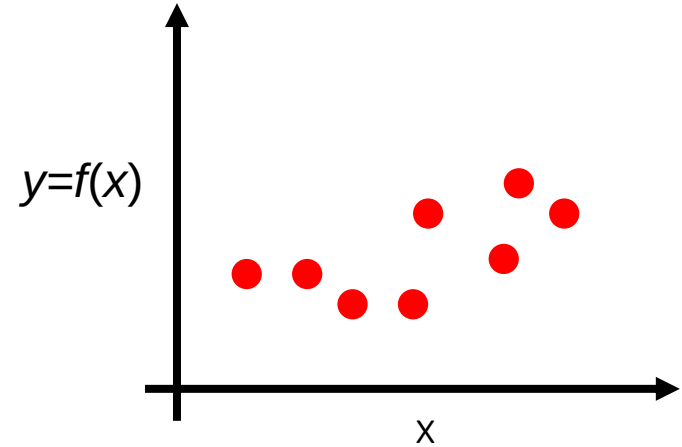
- This is a highly simplified model of real learning:
 - Ignores prior knowledge;
 - Assumes a deterministic, observable “environment”;
 - Assumes examples are given;



Learning is a search through **the space of possible hypotheses $\mathcal{H}$** for one that will perform well, even on new examples beyond the training set

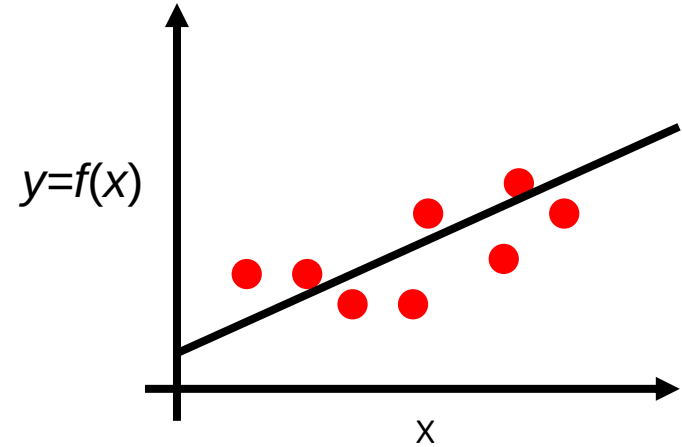
Inductive learning

- Construct/adjust h to agree with f on training set
 - (h is **consistent** if it agrees with f on all examples)
- E.g., curve fitting:



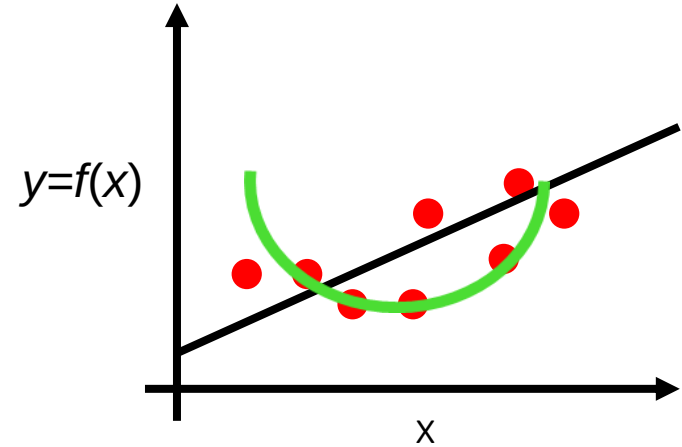
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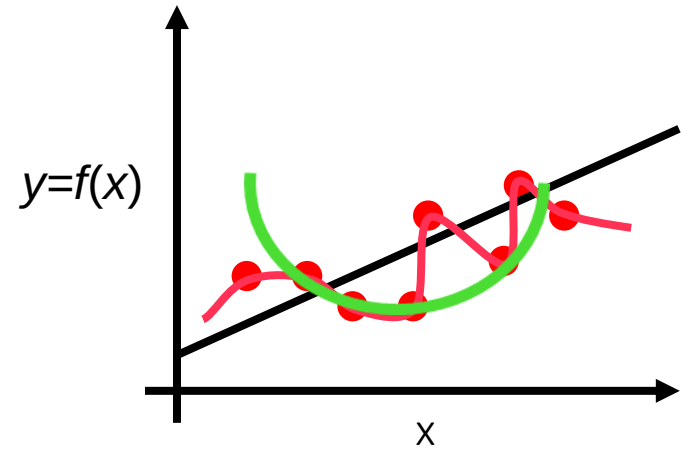
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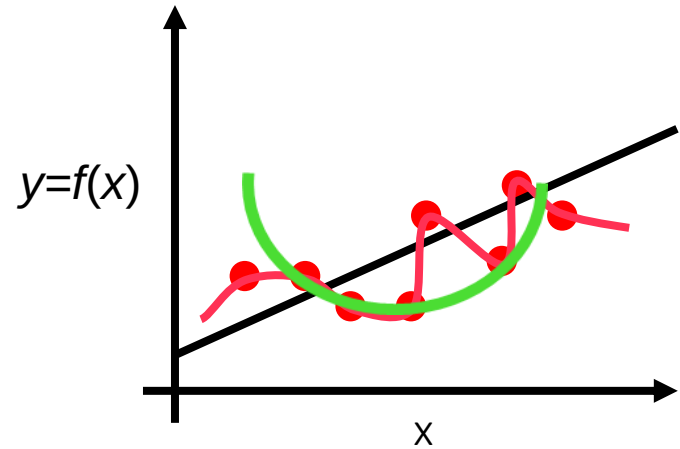
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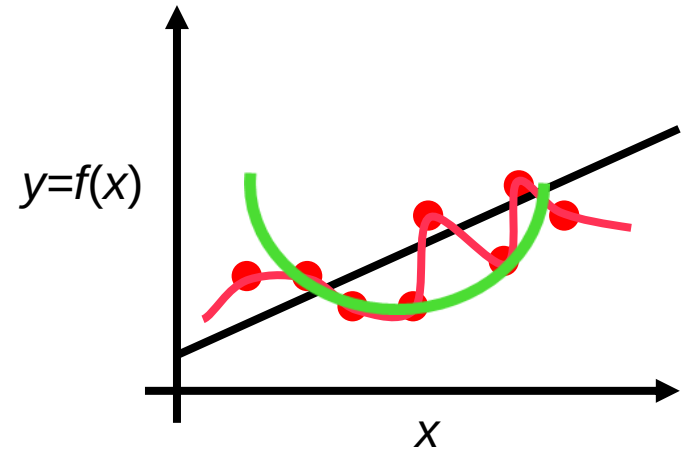
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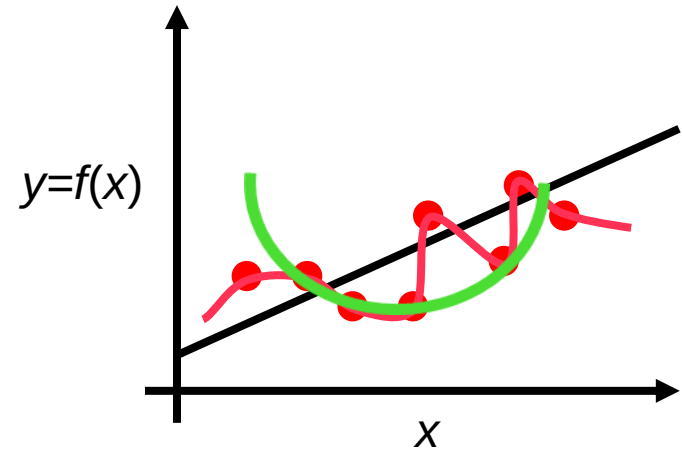
Overfitting

- Learns the “data” and not the underlying function



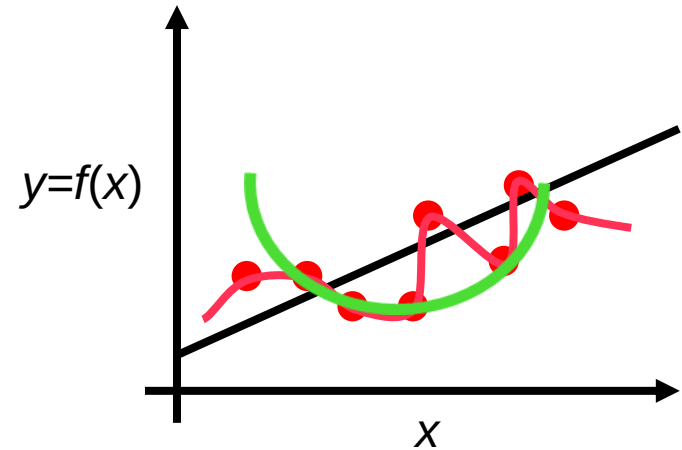
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- Performs well on the data used during the training and poorly with not yet observed data.



Overfitting

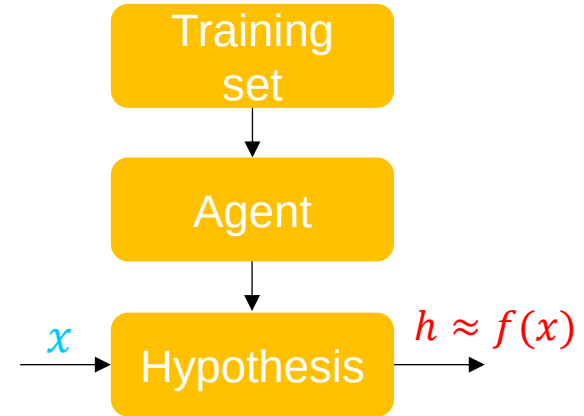
- Learns the “data” and not the underlying function
- Performs well on the data used during the training and poorly with not yet observed data.
- Ockham’s razor:
 - Among competing hypotheses, the one with the fewest assumptions should be selected
 - Prefers the simplest hypothesis consistent with data
 - Maximizes a combination of consistency and simplicity



Hypothesis selection

- Choose the most probably hypothesis given the data

$$\mathbf{h} = \arg \max_{\mathbf{h} \in \mathcal{H}} P(\mathbf{h} \mid \text{data})$$



A hypothesis $\mathbf{h}$ generalizes well if it correctly predicts the value of $\mathbf{y}$ for novel examples

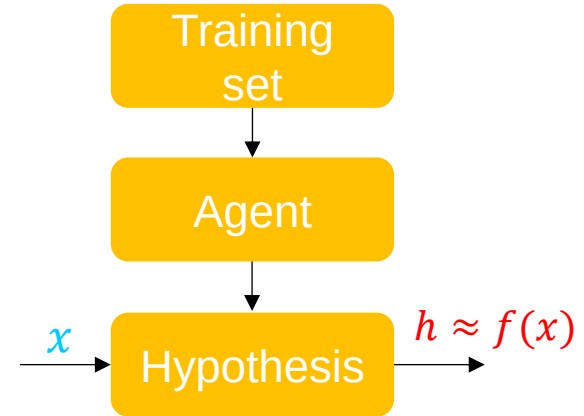
Hypothesis selection

- Choose the most probably hypothesis given the data

$$h = \arg \max_{h \in \mathcal{H}} P(h \mid \text{data})$$

- By Bayes' rule this is equivalent to

$$h = \arg \max_{h \in \mathcal{H}} P(\text{data} \mid h) P(h)$$



A hypothesis h generalizes well if it correctly predicts the value of y for novel examples



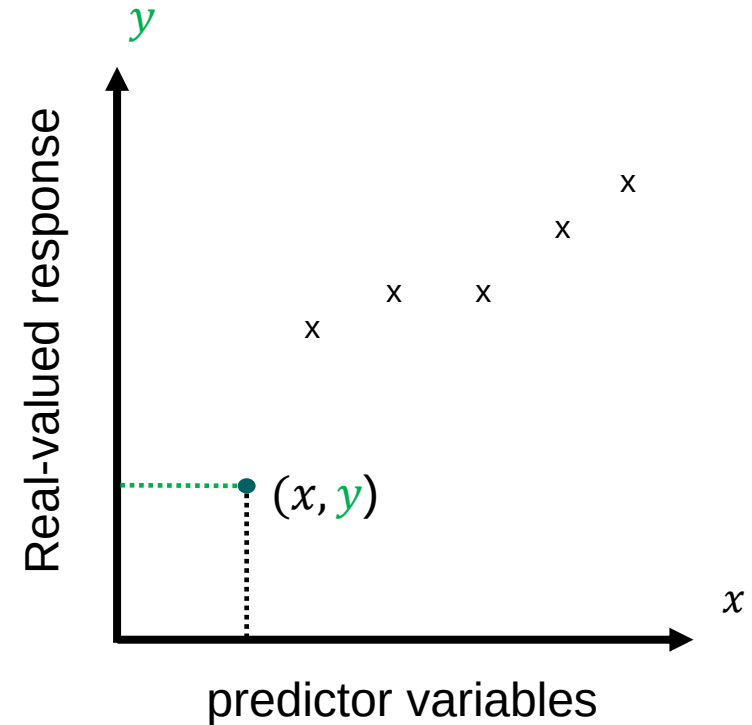
Linear models

Actual value y

- Assume the following functional relationship

$$y = f(x)$$

- y is the actual or observed value

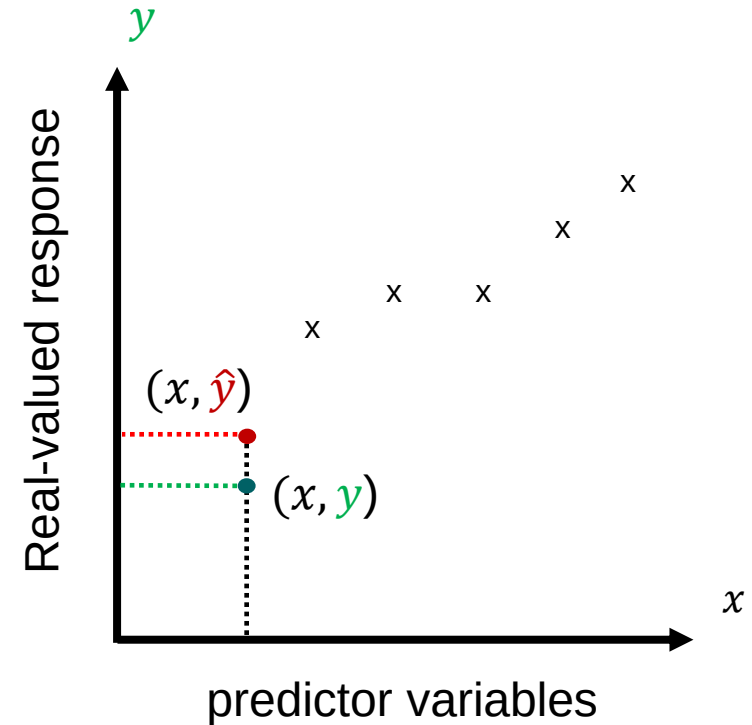


Predicted value $\hat{y}$

- Assume the following functional relationship

$$y = f(x)$$

- y is the actual or observed value
- $\hat{y}$ is a predicted value



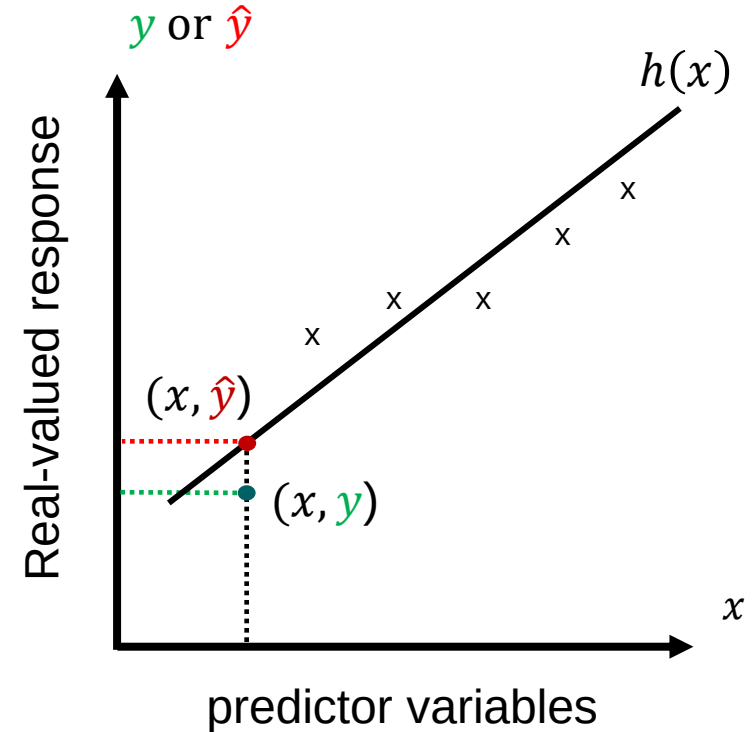
Linear predictor of $\hat{y}$

- Assume the following functional relationship

$$y = f(x)$$

- y is the actual or observed value
- $\hat{y}$ is a predicted value
- $h(x)$ is a linear function that generates a predicted value $\hat{y}$

$$y = f(x) \approx h(x) = \hat{y}$$



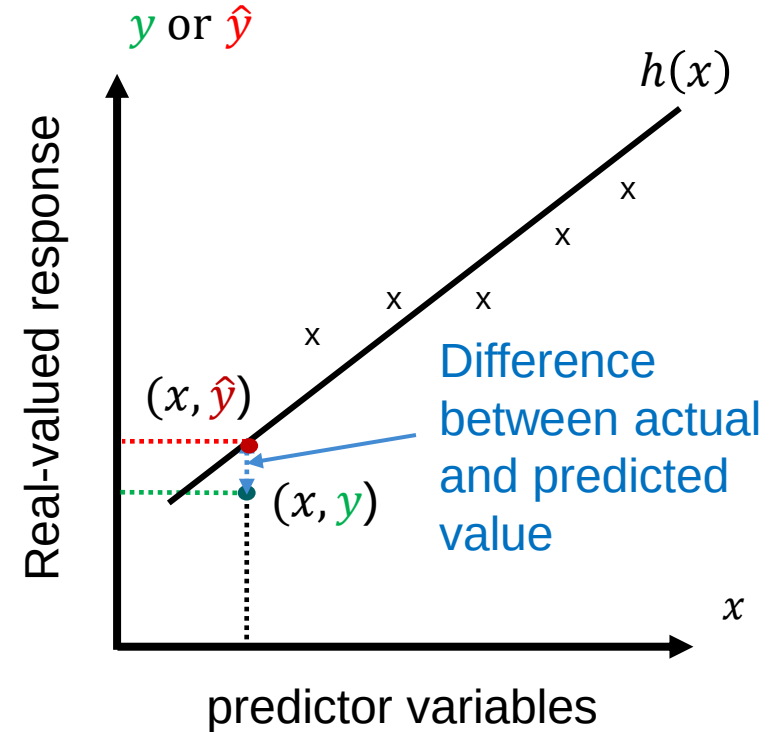
Error model

- Assume the following functional relationship

$$y = f(x) \approx h(x) = \hat{y}$$

- The error or residual is defined as

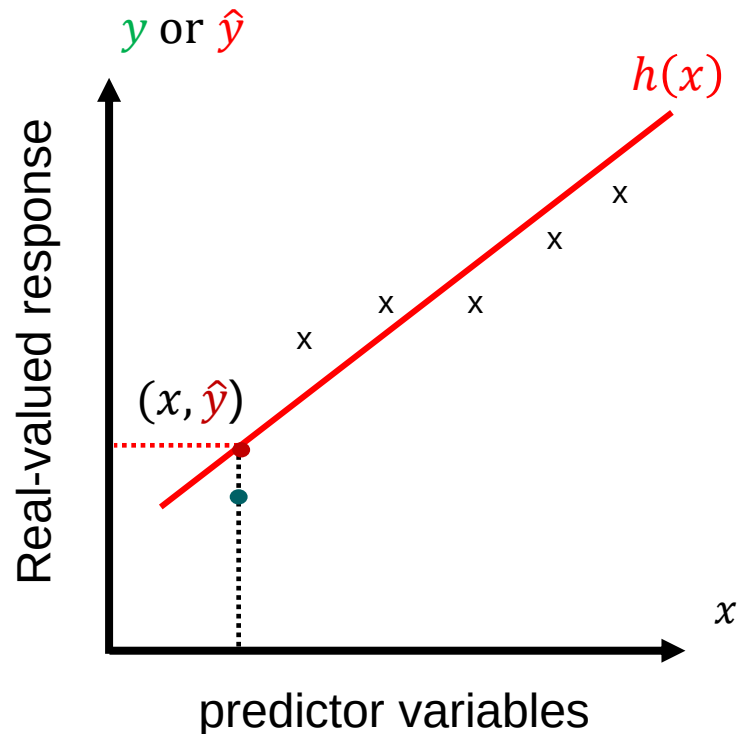
$$\begin{aligned} e &= \hat{y} - y \\ &= h(x) - f(x) \end{aligned}$$



A linear regression model

- Building the model determines the value of output variable $\hat{y}$ as a function of the input (predictor) variables x

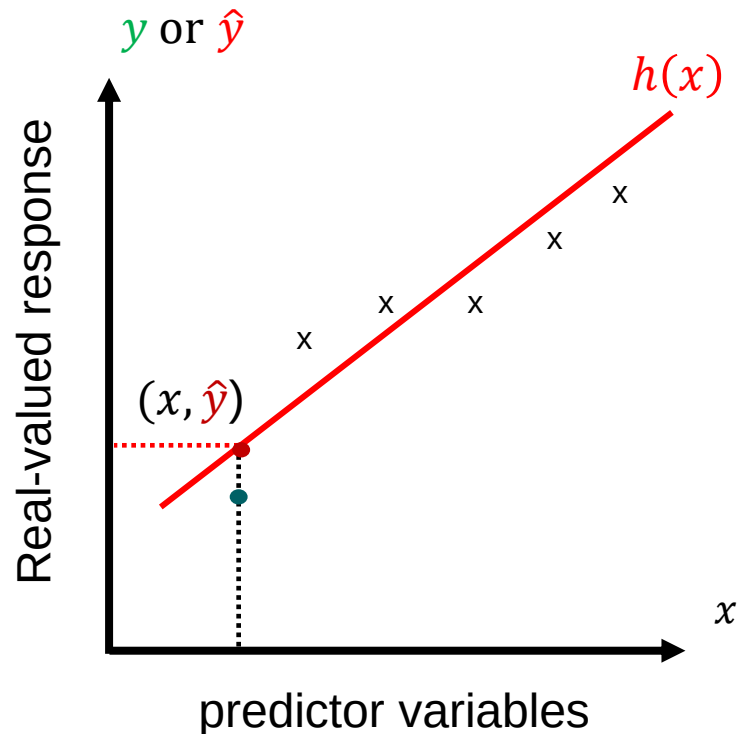
$$\hat{y} = h(x) \approx f(x) = y$$



A linear regression model

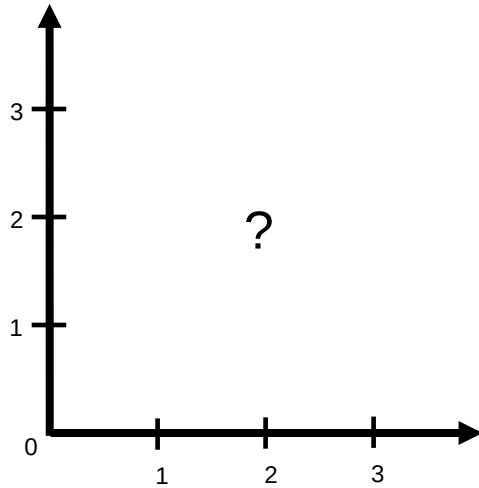
- Building the model determines the value of output variable $\hat{y}$ as a function of the input (predictor) variables x
$$\hat{y} = h(x) \approx f(x) = y$$
- The regression model becomes linear when the unknown function h is a **linear function** of x

Curve fitting problem!



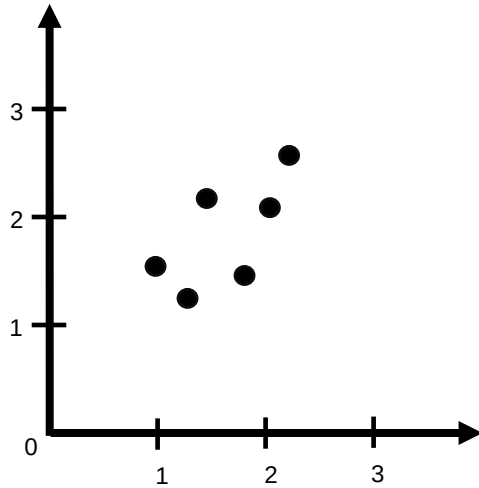
Examples of linear models

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$



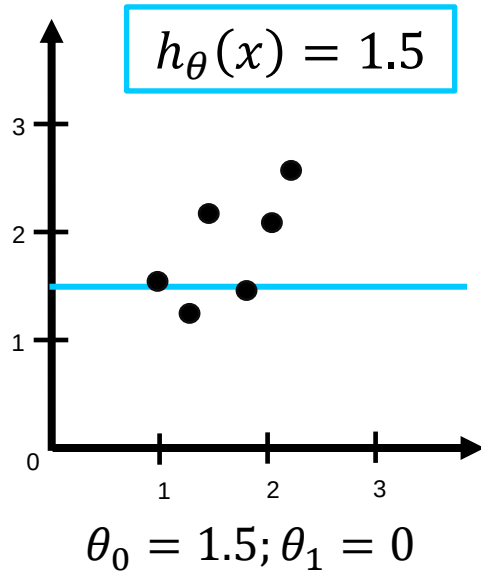
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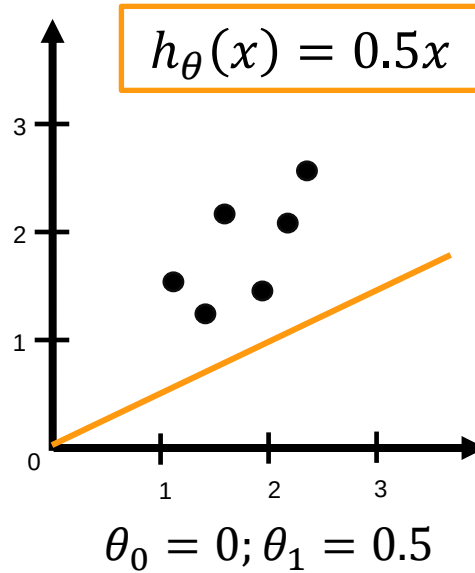
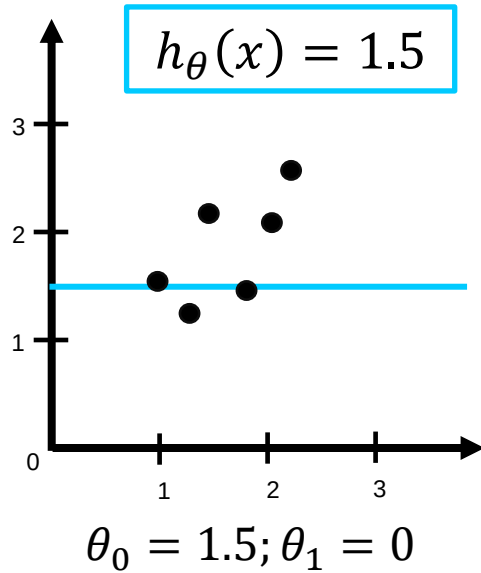
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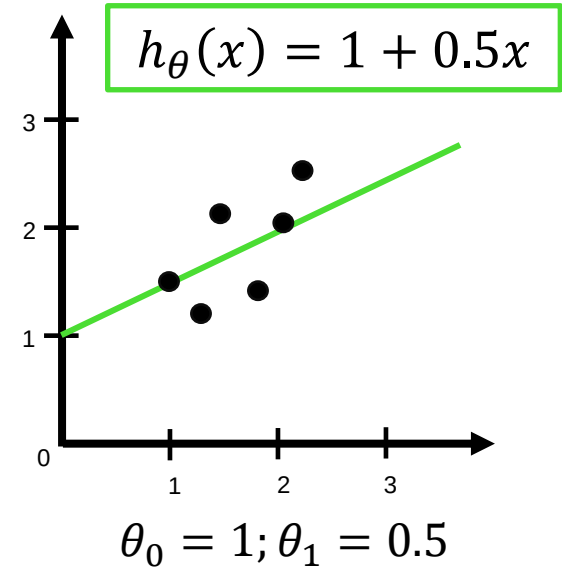
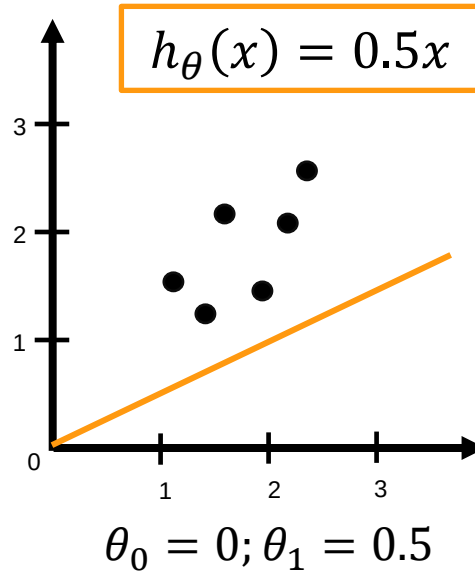
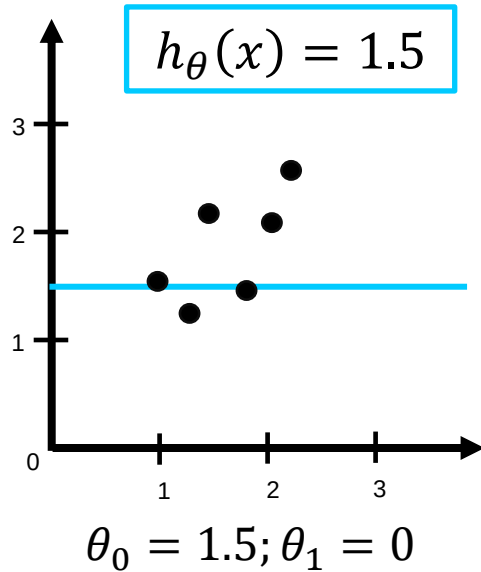
Examples of linear models

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Examples of linear models

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Dataset and data pre-processing

Datasets

- **Training set:**

- a set of examples used for learning, where the target value is known.

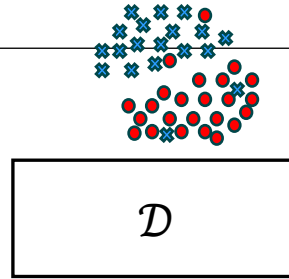
- **Validation set:**

- a set of examples used to tune the architecture of a classifier and estimate the error

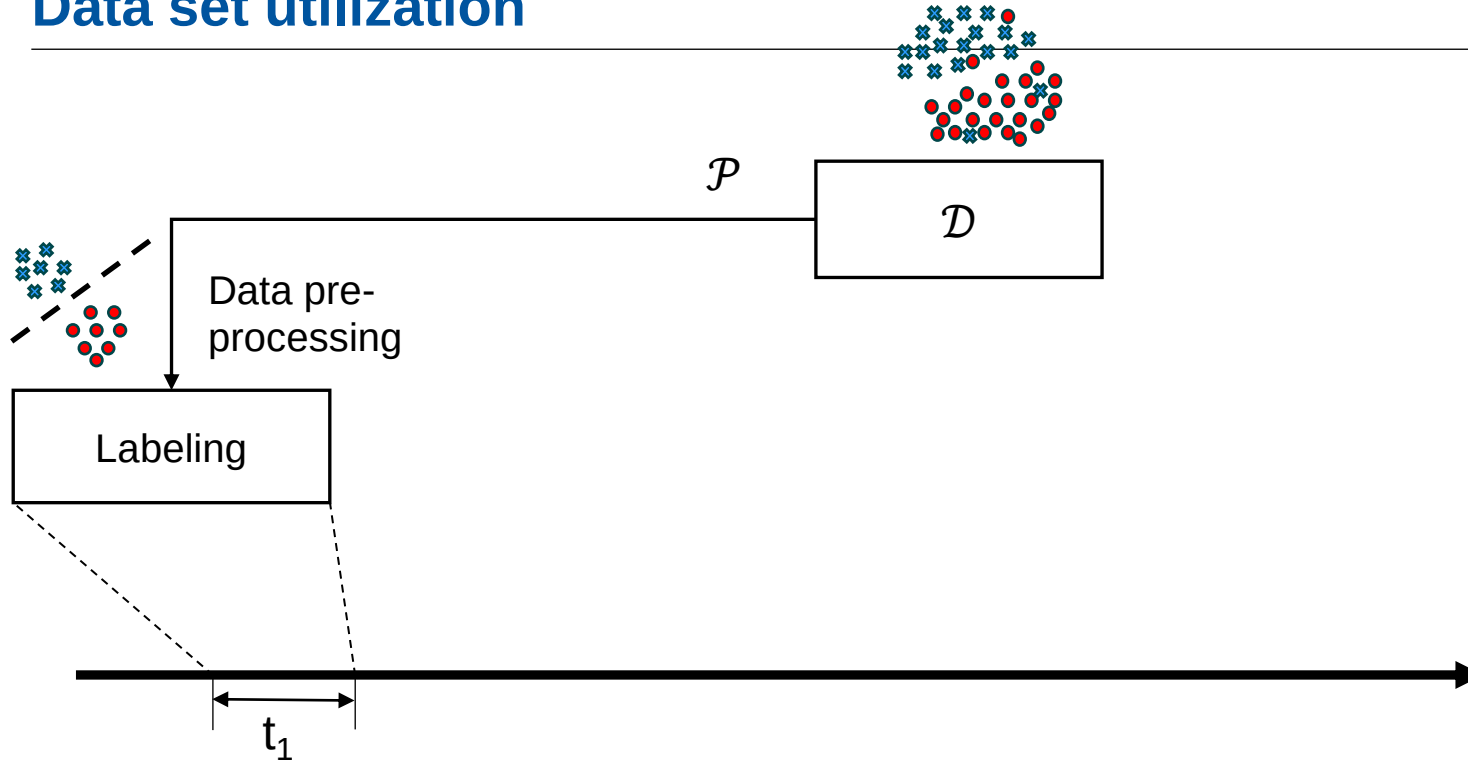
- **Testing set:**

- used only to assess the achievable performance of a classifier.
- **It is never used** during the training process so that the error on the prediction provides an unbiased estimate of the generalization error

Data set utilization

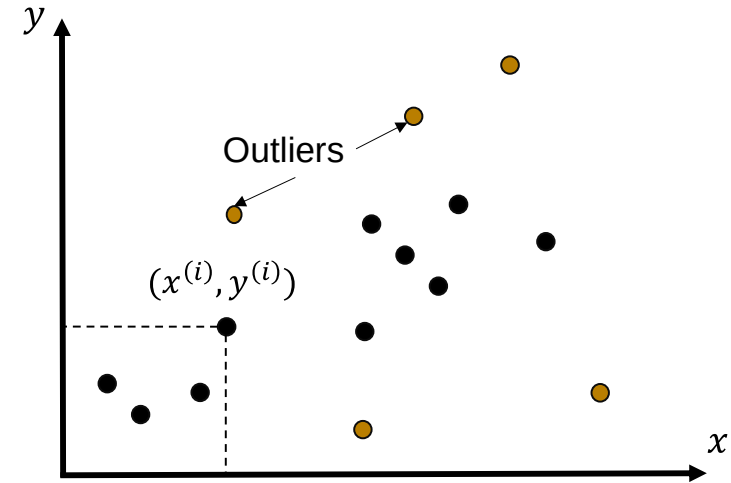


Data set utilization



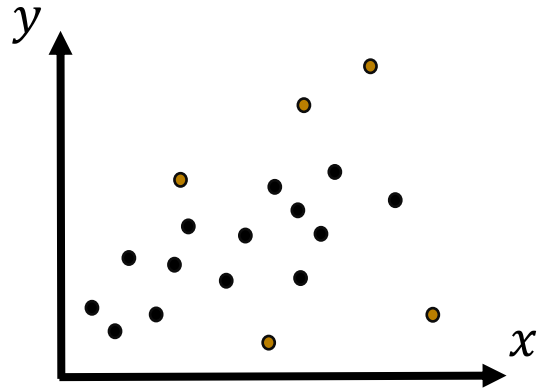
Pre-processing data tasks

- Data labeling
- Data sampling to select a representative subset
- De-noising
- Handle missing data
- Data normalization
- Extract specific data in some particular context
 - Feature extraction (unsupervised learning)



Preprocessing of data

Raw dataset with outliers

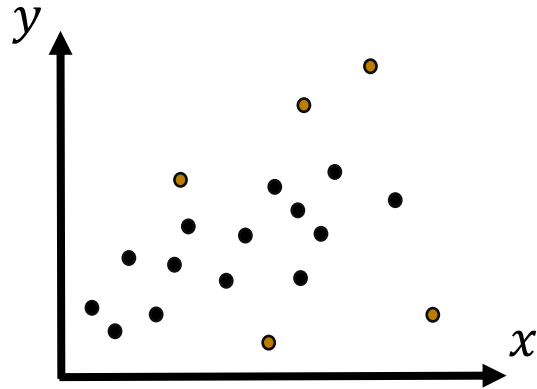


$n > m$ data points

Preprocessing of data

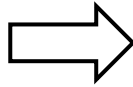
Training set: $\{x^{(i)}, y^{(i)}\}_{i=1}^m$

Raw dataset with outliers

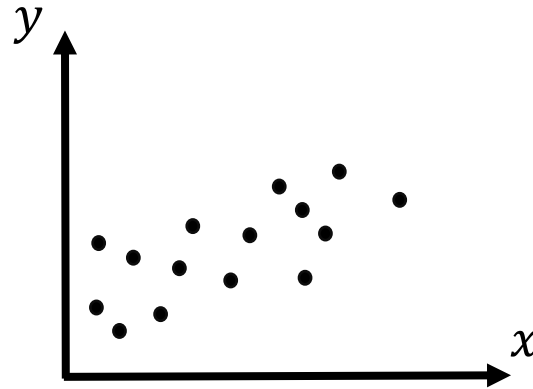


$n > m$ data points

Preprocessing
to remove
outliers

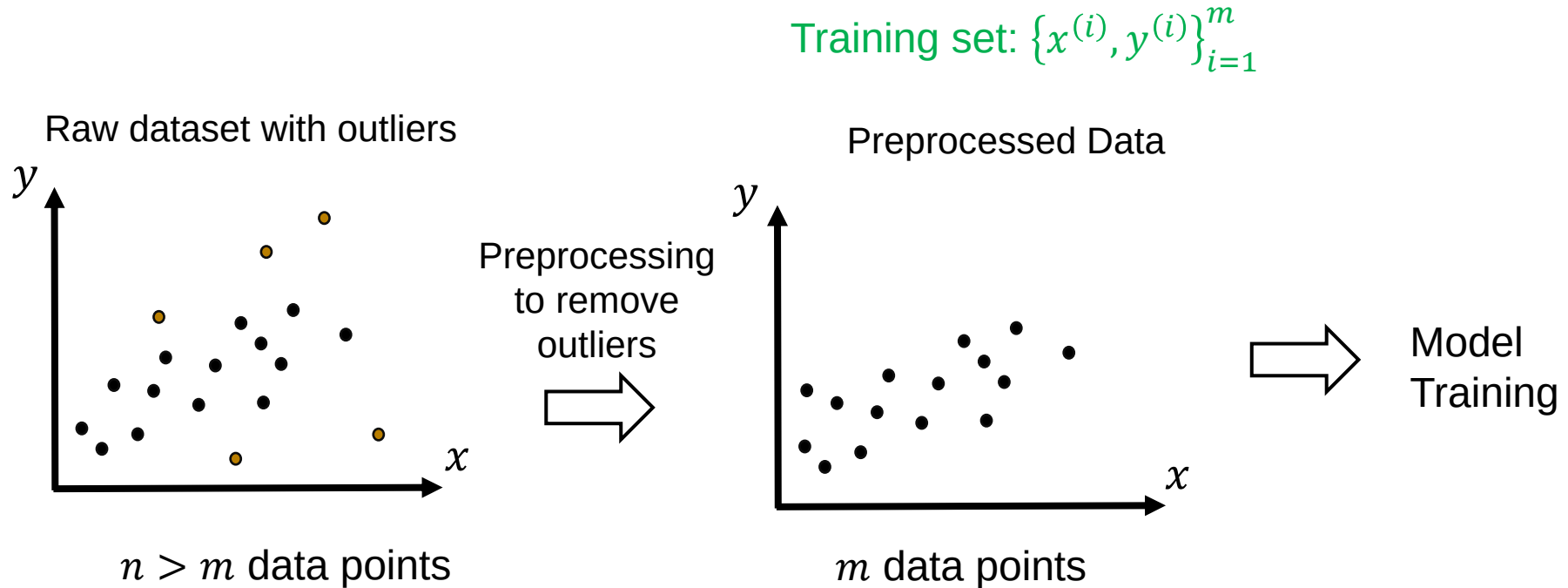


Preprocessed Data



m data points

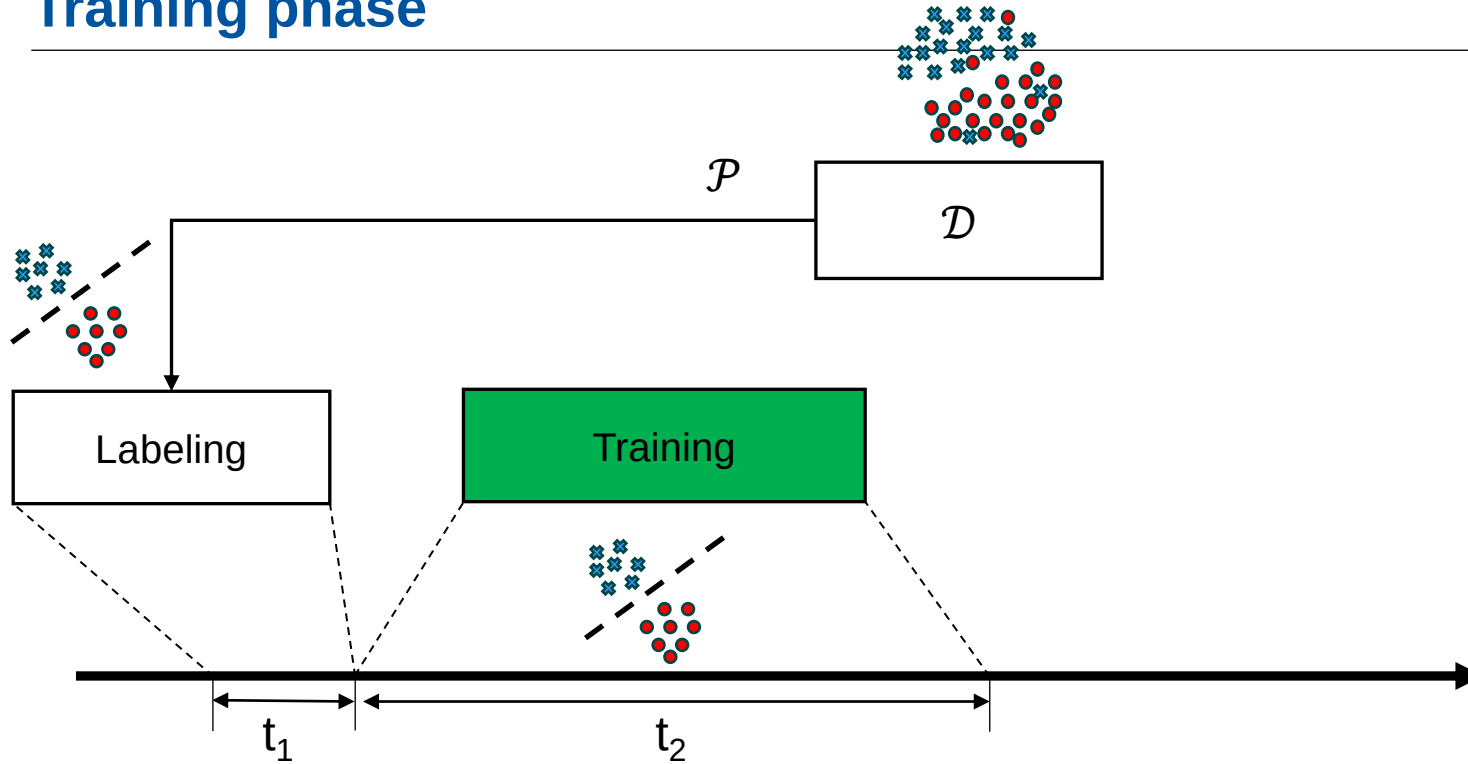
Preprocessing of data





Model training

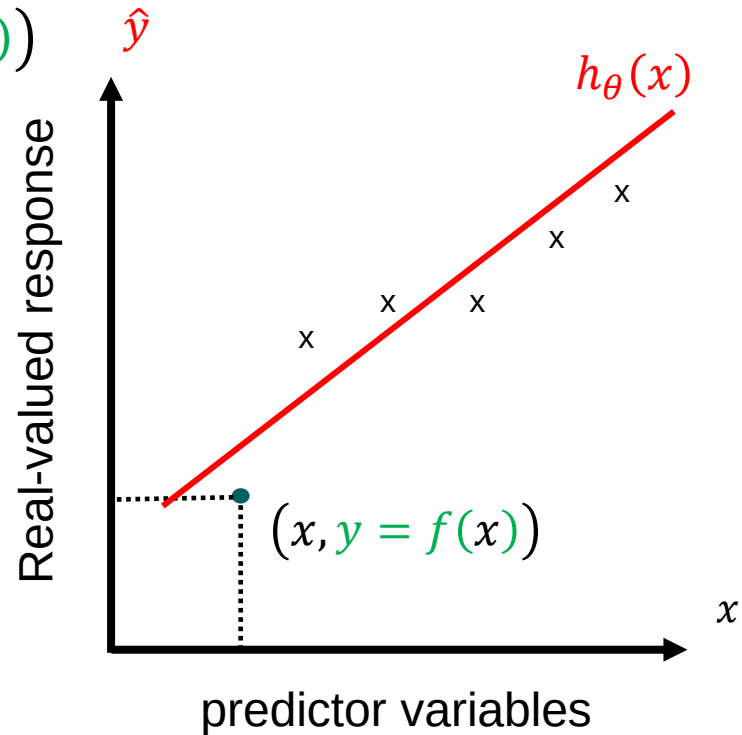
Training phase



Training for single data point (x, y)

- We are given a training point $(x, y = f(x))$ and a linear model

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$



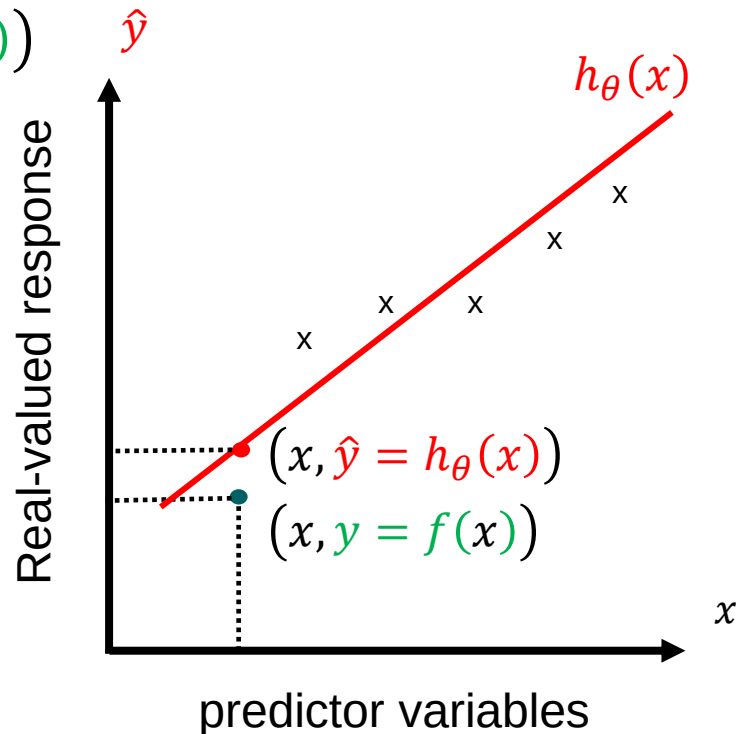
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$$h_{\theta}(x) \approx y$$



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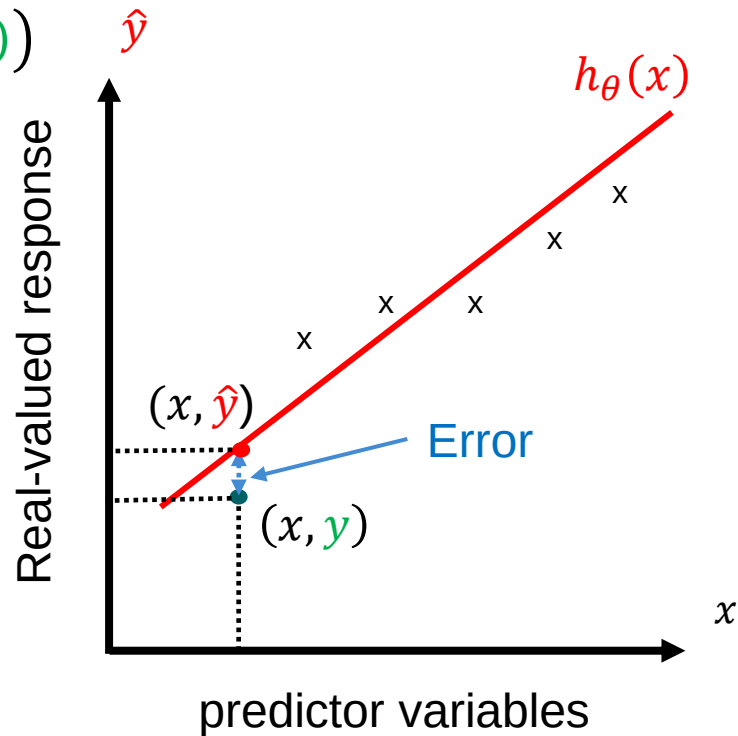
$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

- Select θ_0 and θ_1 so that

$$h_{\theta}(x) \approx y$$

while minimizing the error:

$$e = \hat{y} - y = h_{\theta}(x) - y$$



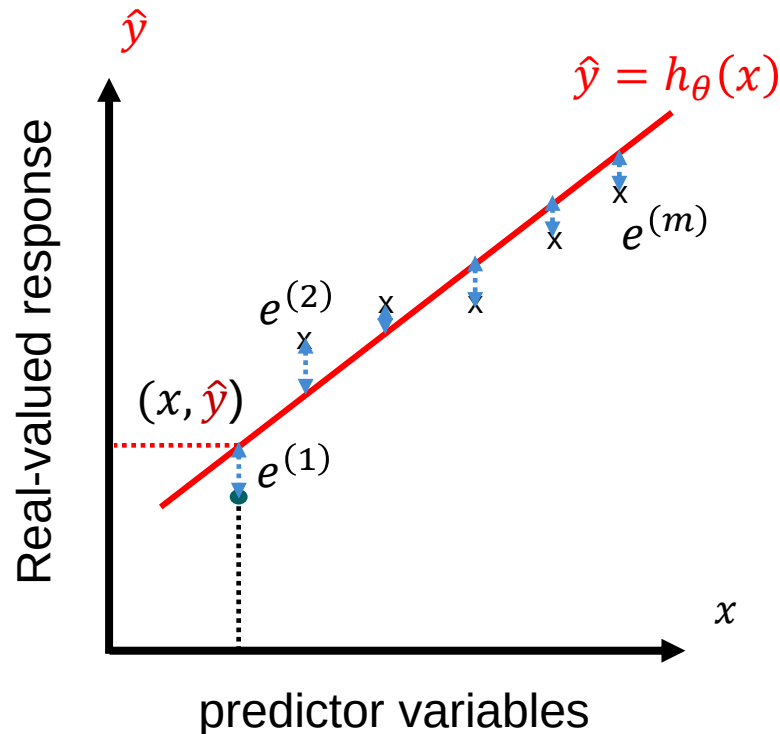
Training for m data sets

- For a given m training examples $\{(x^{(i)}, y^{(i)})\}_{i=1}^m$ and a linear model

$$h_{\theta}(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$$

- Select θ_0 and θ_1 so that $h_{\theta}(x^{(i)}) \approx y^{(i)}$, while minimizing the total error:

$$e = \sum_i e^{(i)} = \sum_i (h_{\theta}(x^{(i)}) - y^{(i)})$$



Training for m data sets

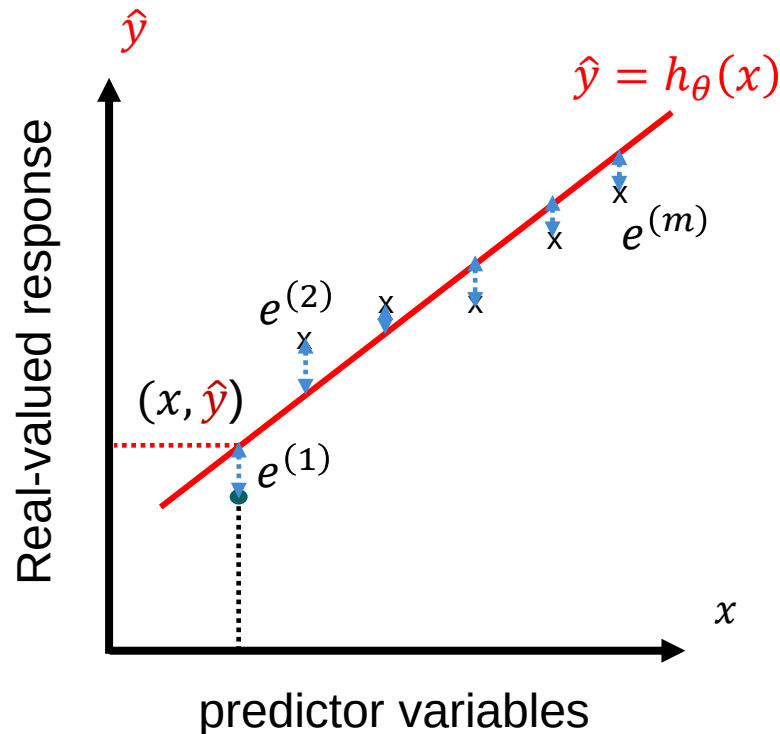
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How to minimize the error?



Cost function

Initializing model parameters

① Initialization
of parameters

$$\theta_0 = 0 \quad \theta_1 = 0.5$$

Initializing hypothesis

①

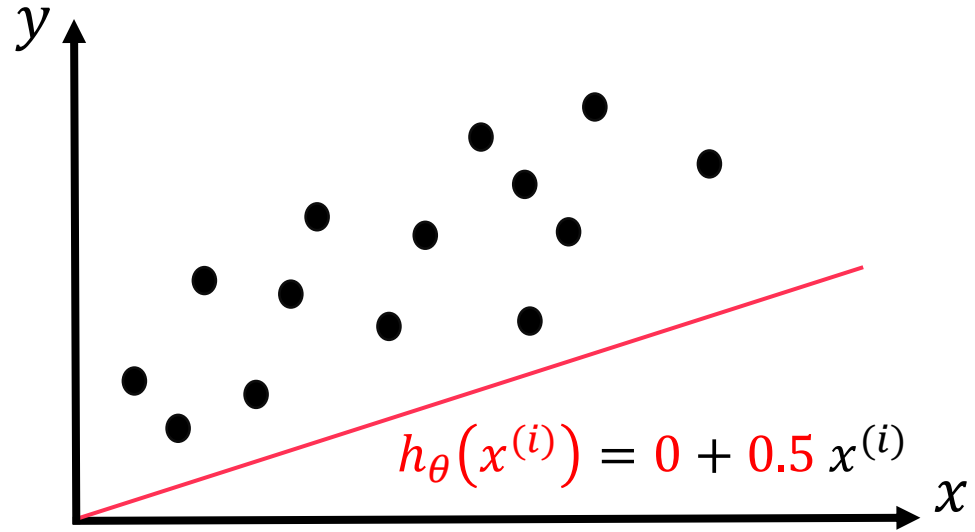
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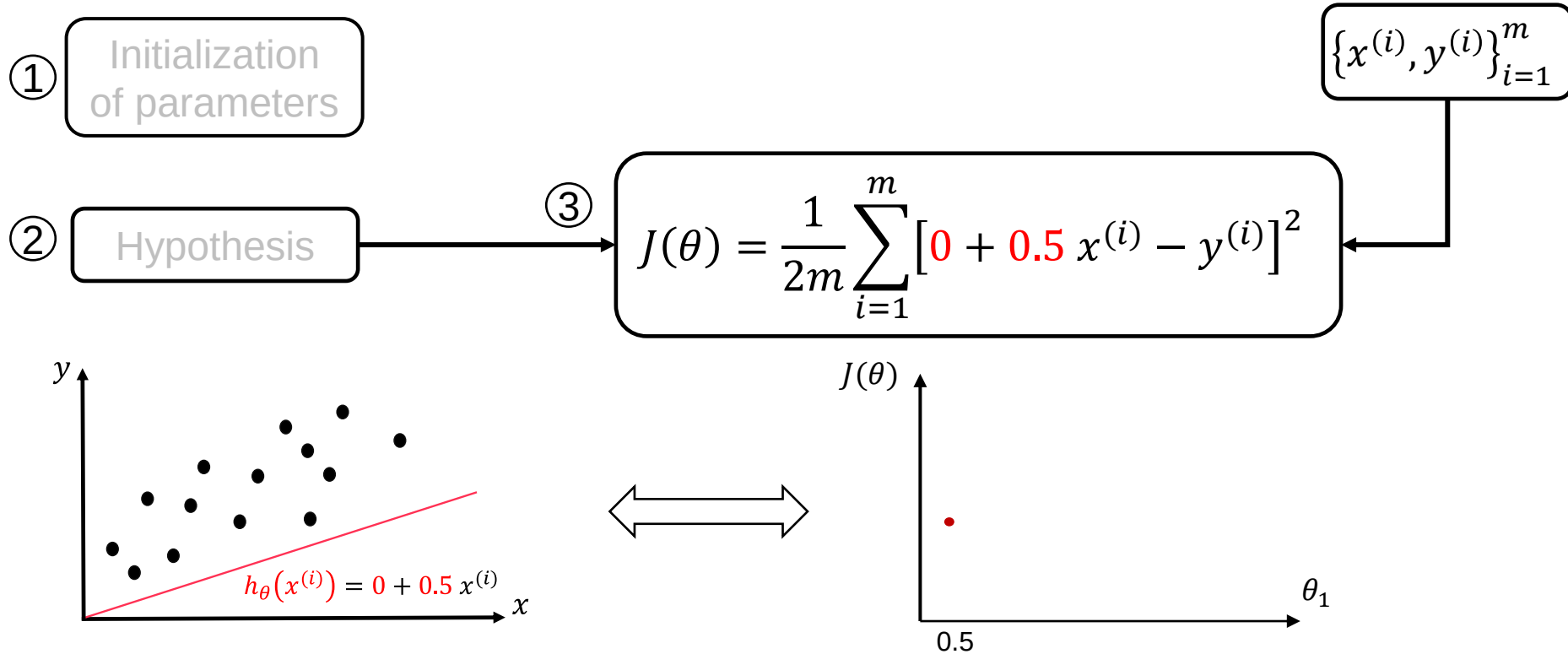
②

Hypothesis

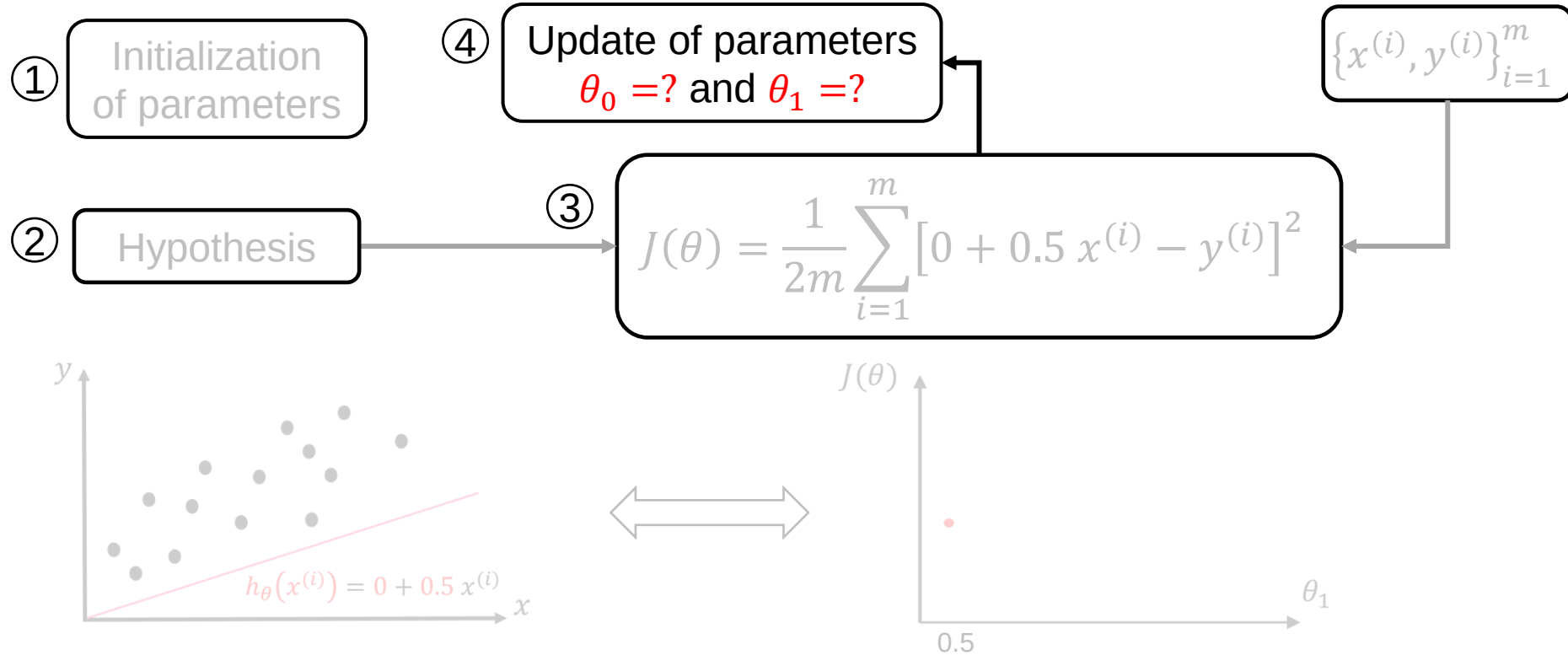
$$\begin{aligned} h_{\theta}(x^{(i)}) &= \theta_0 + \theta_1 x^{(i)} \\ &= 0 + 0.5 x^{(i)} \end{aligned}$$



Computation of the cost function $J(\theta)$



Update of parameters θ_0 and θ_1



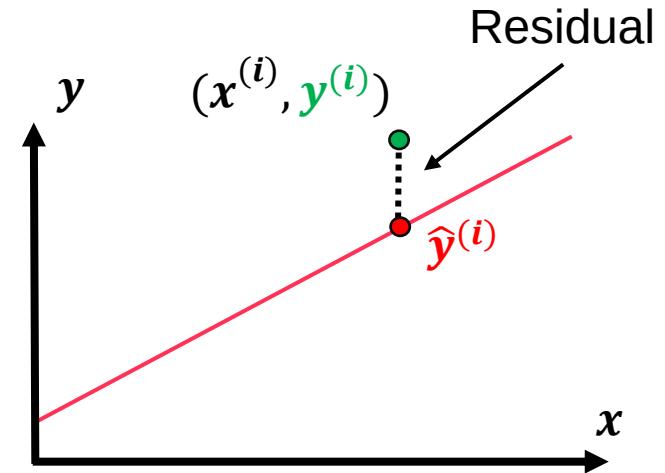
How to update the parameters?

Basic idea

- We search for the most optimal values of θ based on the **least square** method

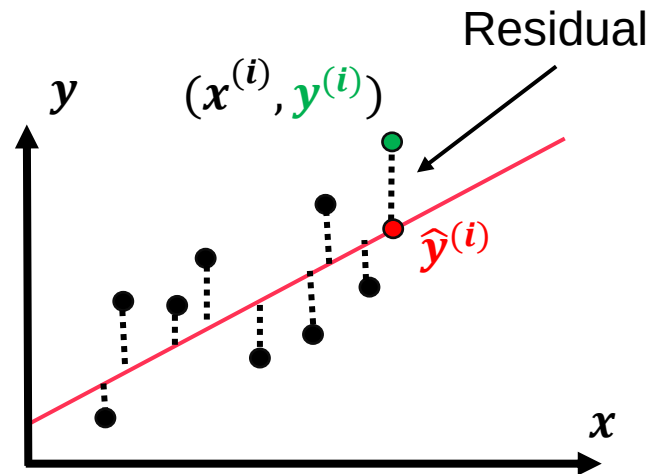
Basic idea

- We search for the most optimal values of θ based on the **least square** method
- The method **searches** for the best linear fit by minimizing the sum of the squared residuals between
 - Predicted output $(x^{(i)}, \hat{y}^{(i)})$, and
 - the actual values $(x^{(i)}, y^{(i)})$



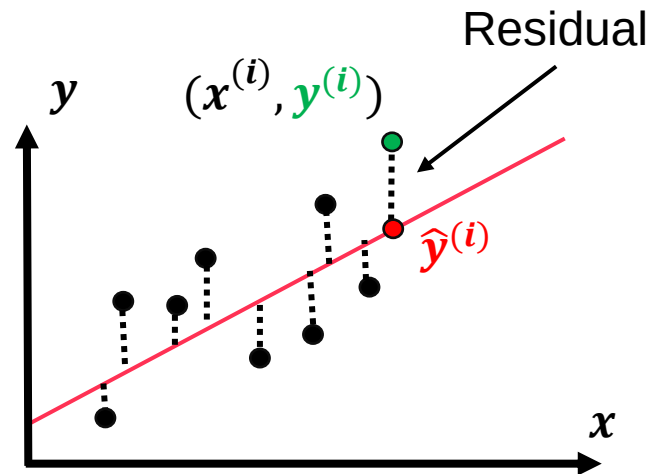
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 - Predicted output $(x^{(i)}, \hat{y}^{(i)})$, and
 - the actual values $(x^{(i)}, y^{(i)})$
- For all of data points in a training set
- For example, the search returns:
④ $\theta_0 = 0 \quad \theta_1 = 0.75$



We go back to the training process...

...Update of hypothesis

④

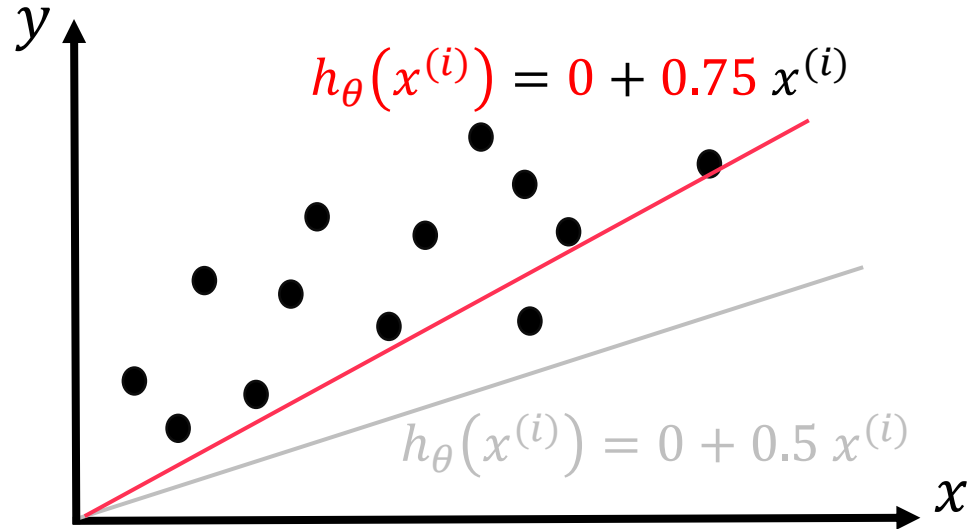
Update of
parameters

$$\theta_0 = 0 \quad \theta_1 = 0.75$$

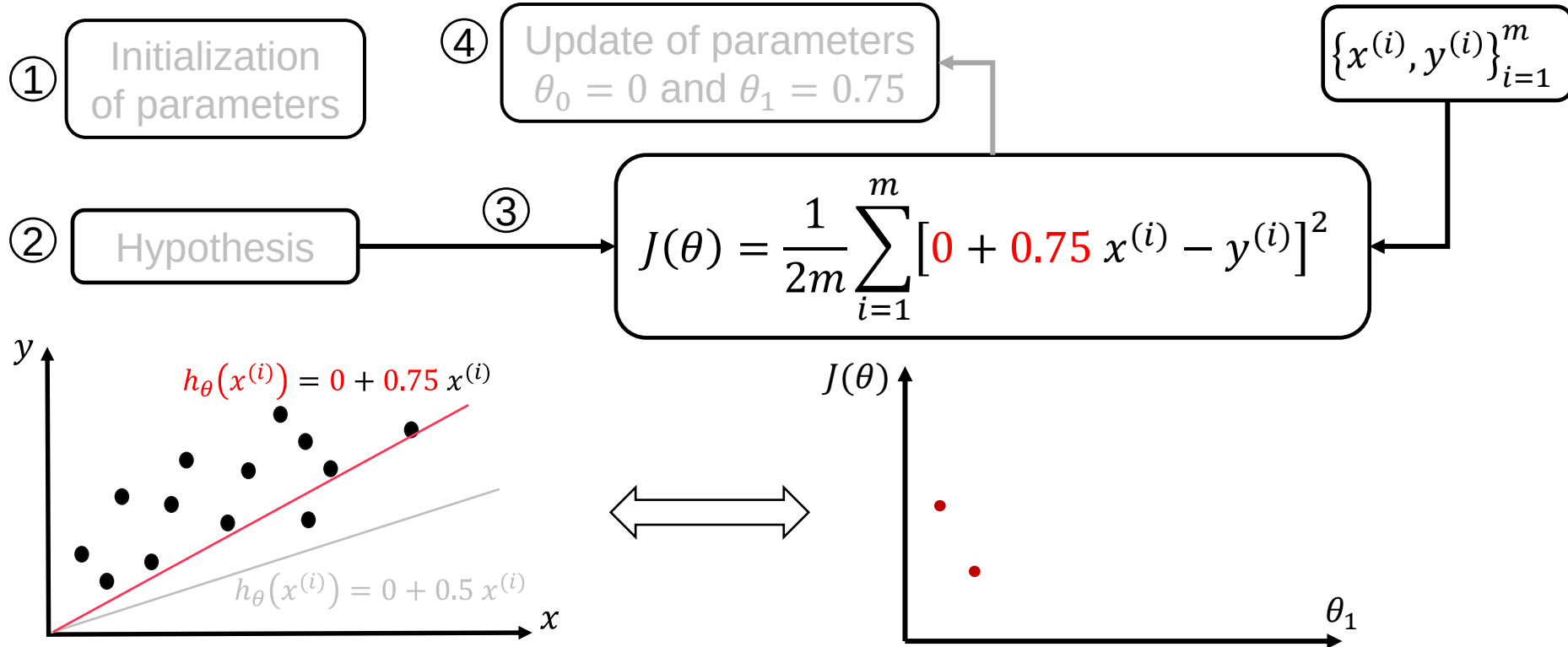
②

Hypothesis

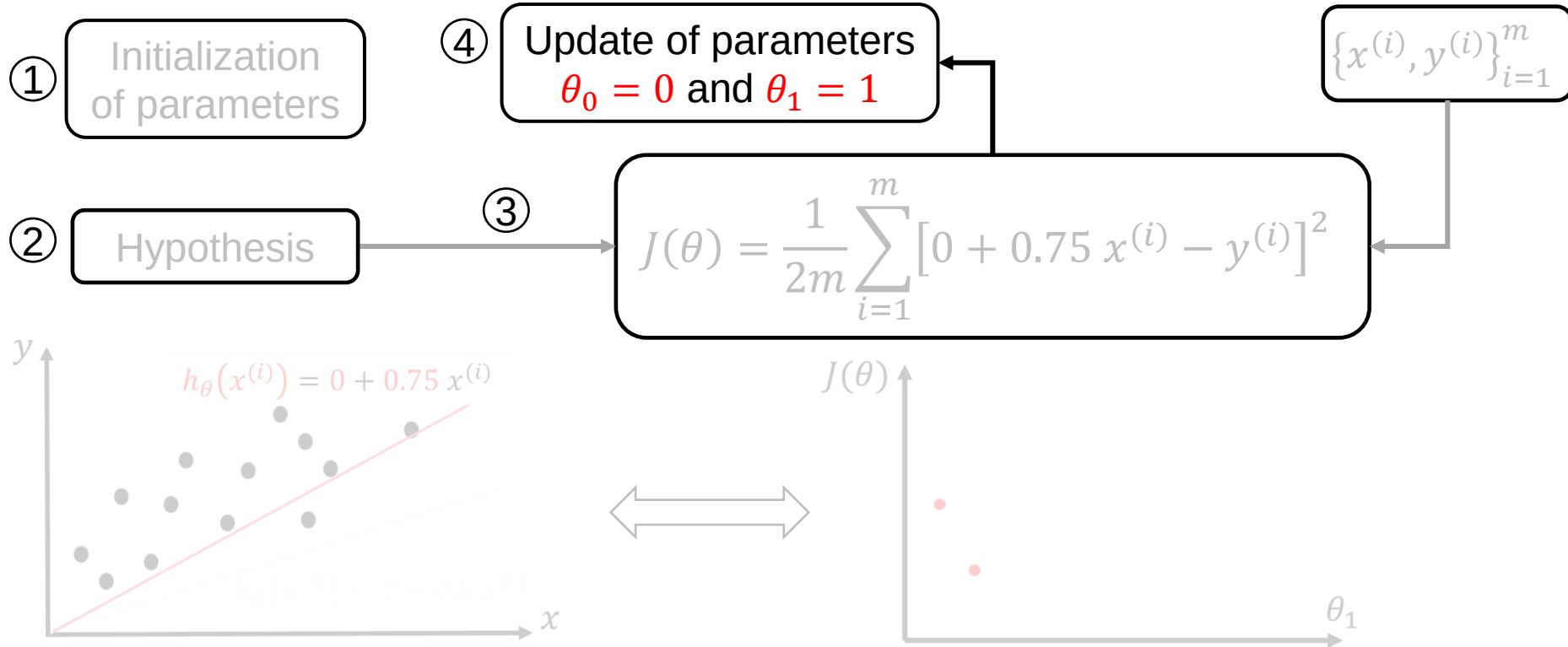
$$\begin{aligned} h_{\theta}(x^{(i)}) &= \theta_0 + \theta_1 x^{(i)} \\ &= 0 + 0.75 x^{(i)} \end{aligned}$$



Computation of the cost function $J(\theta)$



Update of parameters θ_0 and θ_1



Update of hypothesis

④

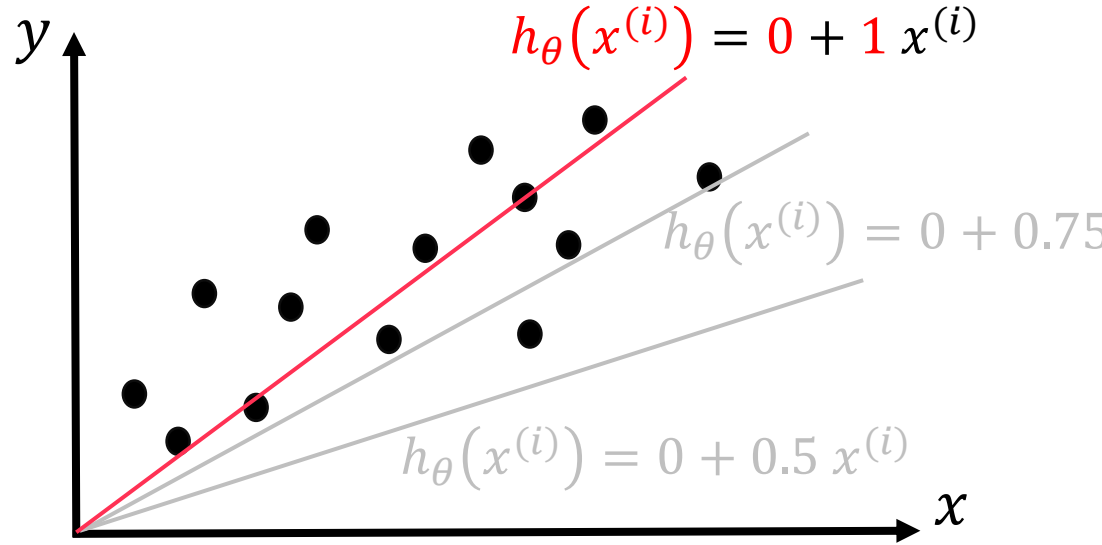
Update of
parameters

$$\theta_0 = 0 \quad \theta_1 = 1$$

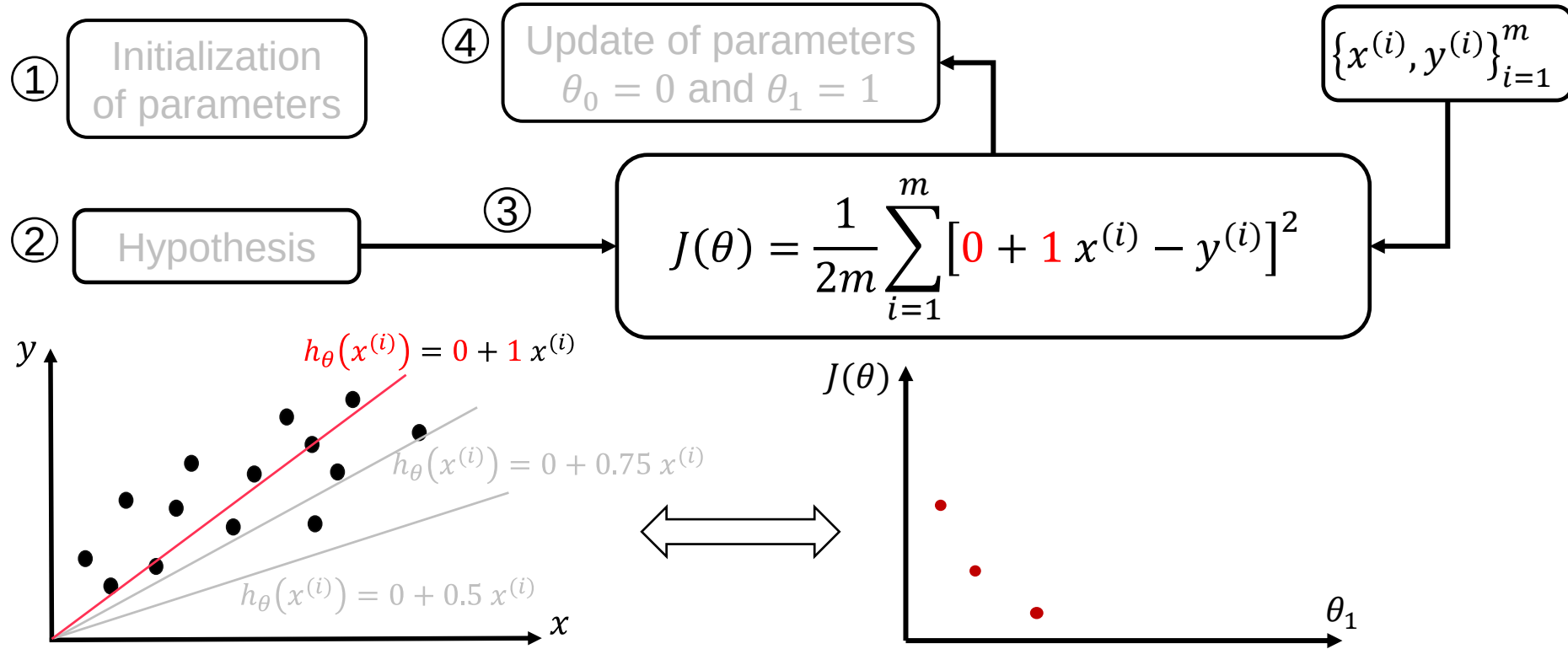
②

Hypothesis

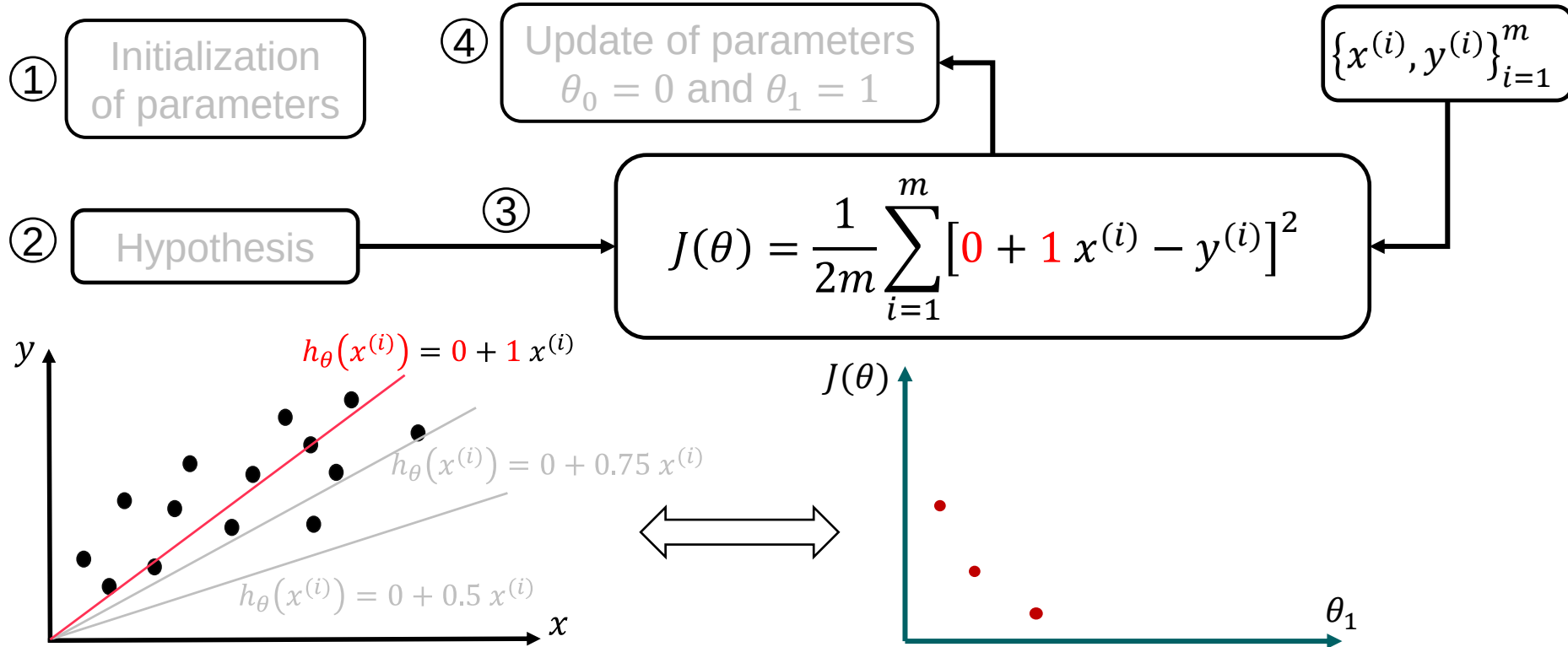
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Update of parameters θ_0 and θ_1



Computation of the cost function $J(\theta)$



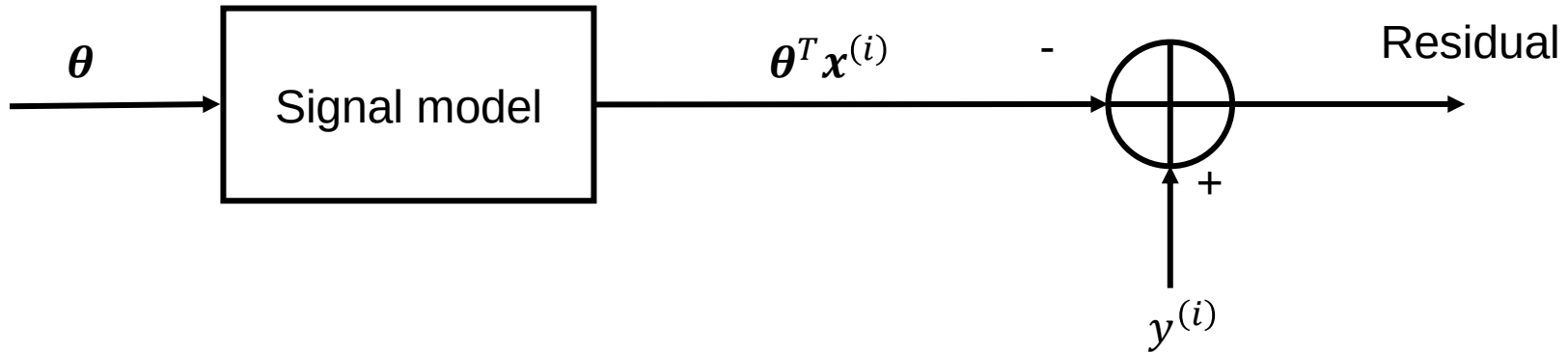
After multiple iterations our training is done...

- When $J(\theta)$ reaches “small” predefined value depending on the application
 - e.g., 10^{-6}

Least-squares computation method

- The **least-squares method** searches for the best fit of a data points by minimizing the sum of the squared residuals

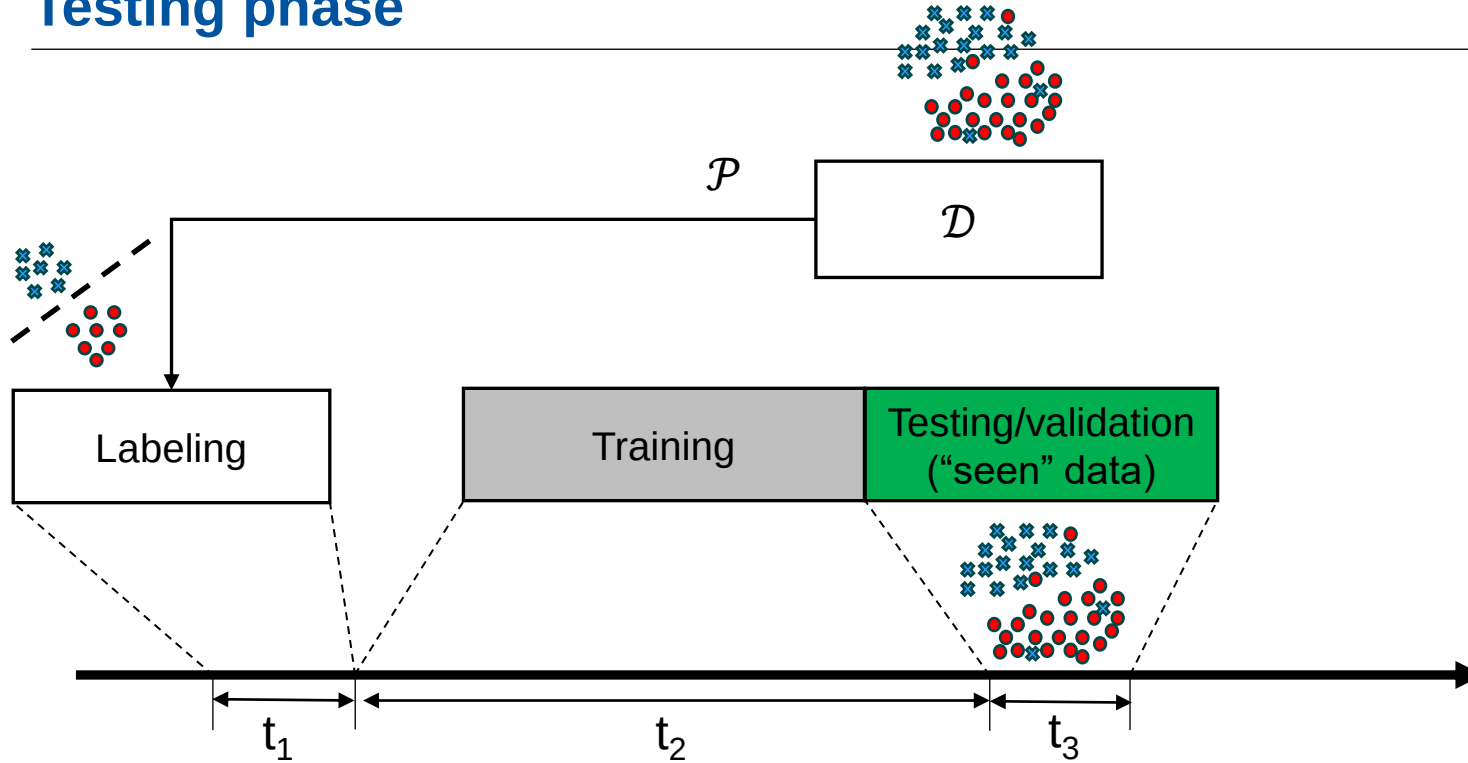
$$\sum_{i=1}^m (y^{(i)} - \boldsymbol{\theta}^T \mathbf{x}^{(i)})^2$$



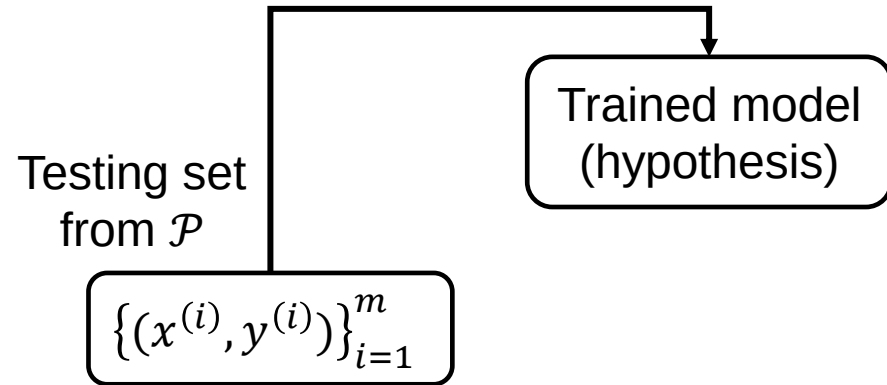


Testing a model on seen data

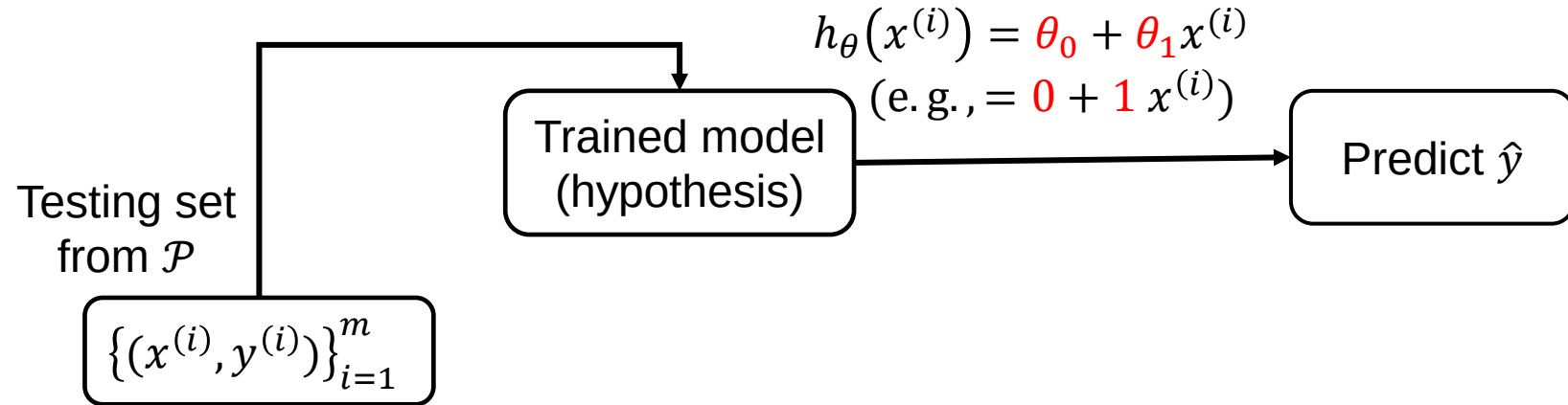
Testing phase



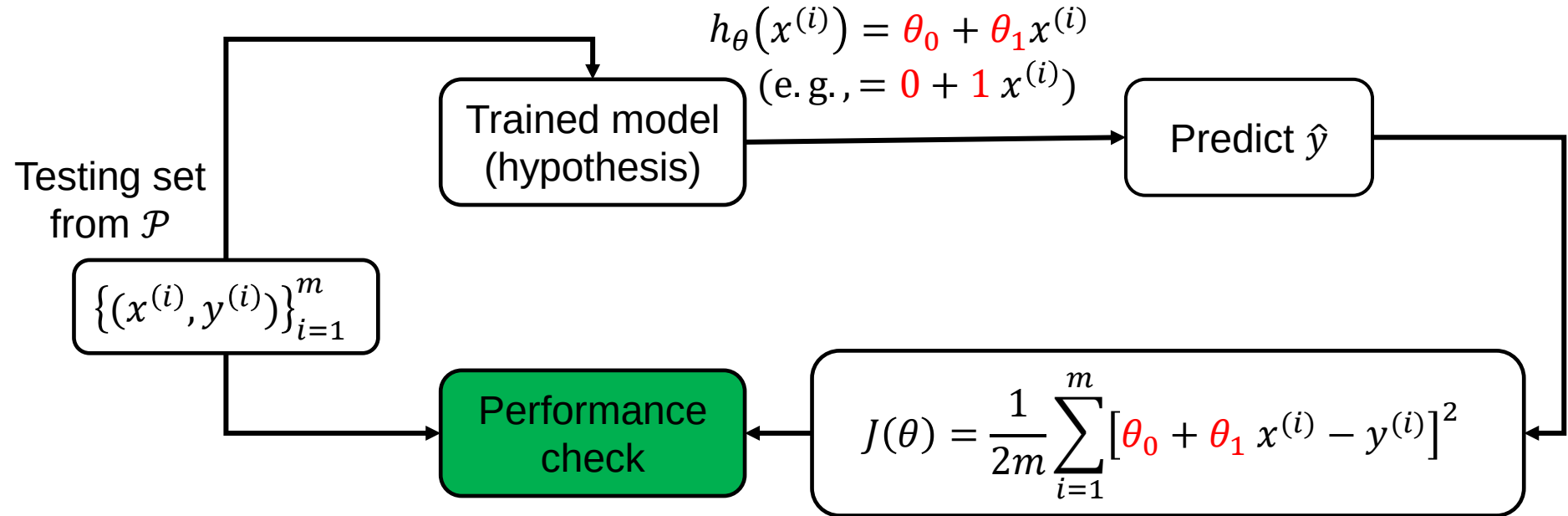
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Testing phase



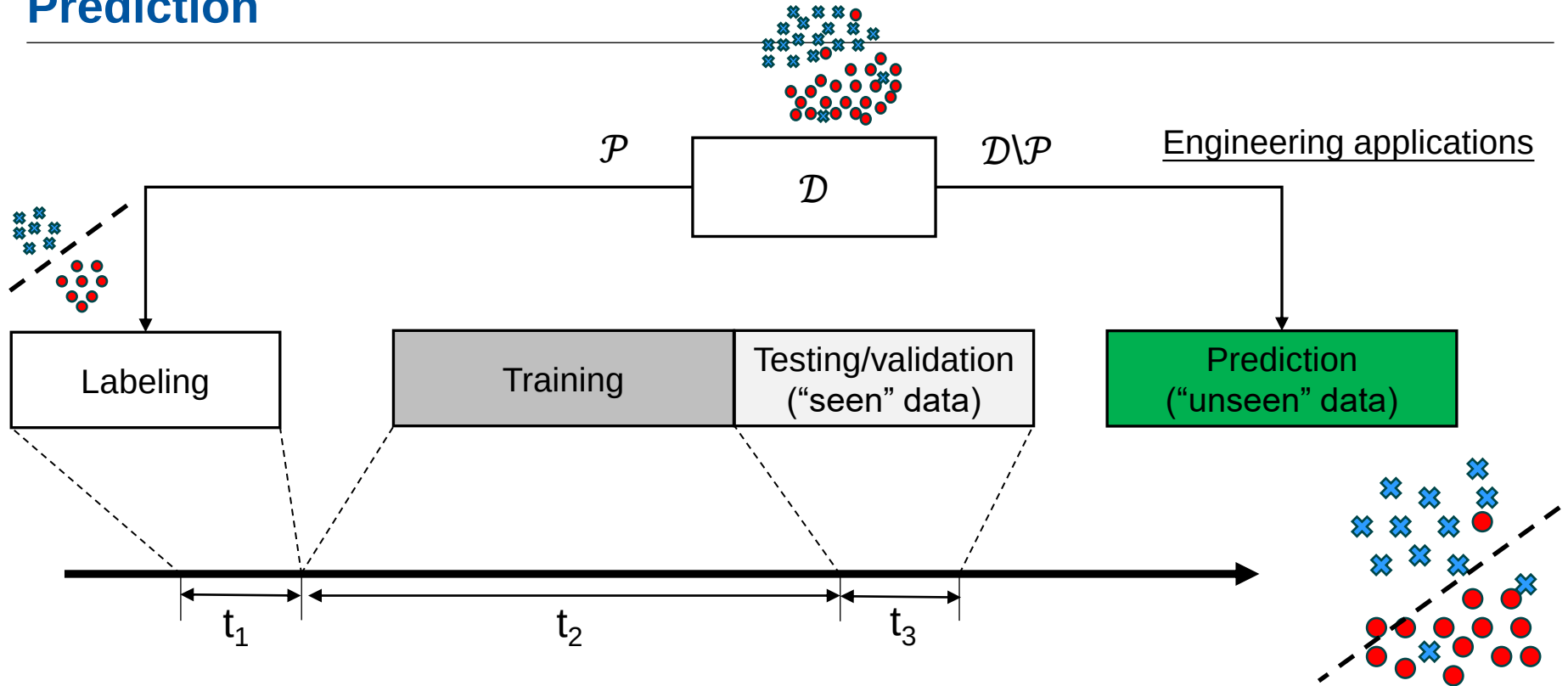
Testing phase



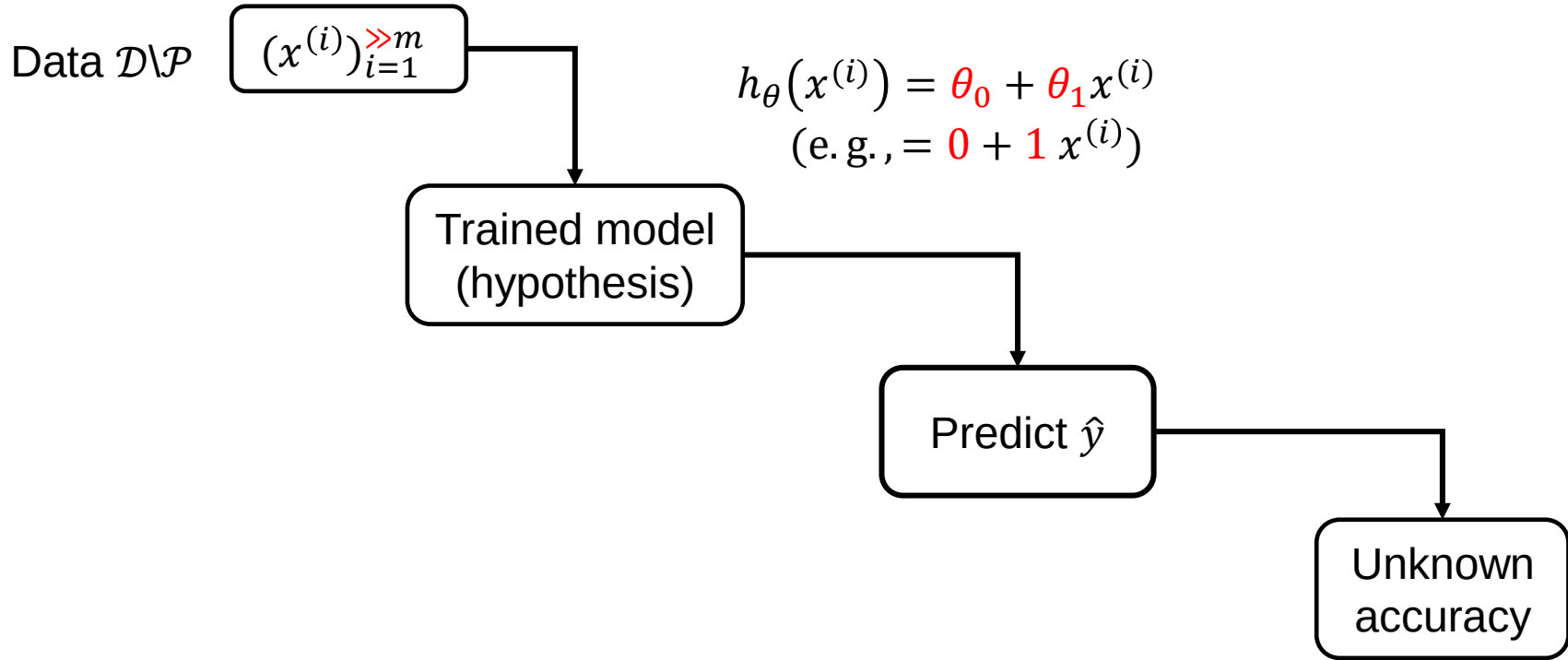


Predicting a model on unseen data

Prediction



Prediction phase

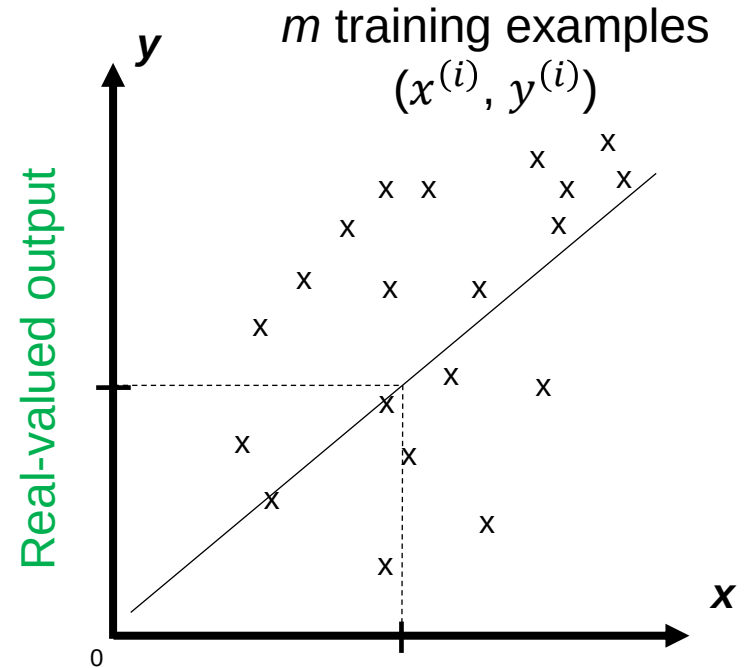




Final remarks

Regression models in supervised learning

- Labeled data in supervised learning
 - the “right answer” for each training example in the data
- Regression problem
 - Predicts **real-valued** output
- Classification problem
 - Predicts **discrete-valued** output

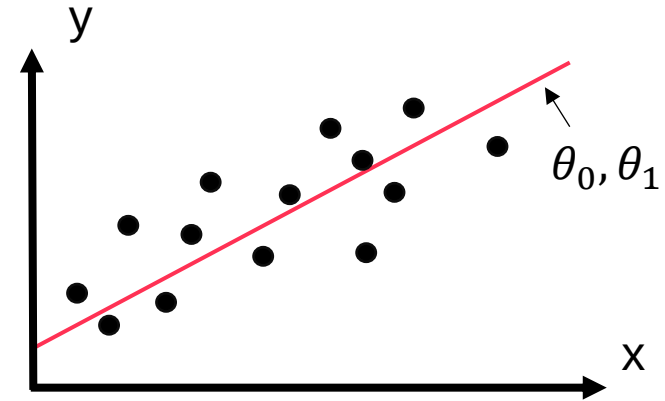


Representation of h with single variable

- Linear regression with one variable (univariate linear regression)

$$\hat{y} = h_{\theta}(x) = \theta_0 + \theta_1 x_1$$

- $x_0 = 1$
- θ are called parameters of a model
- θ defines a slope of the red curve
- $\hat{y} \rightarrow$ predicted value

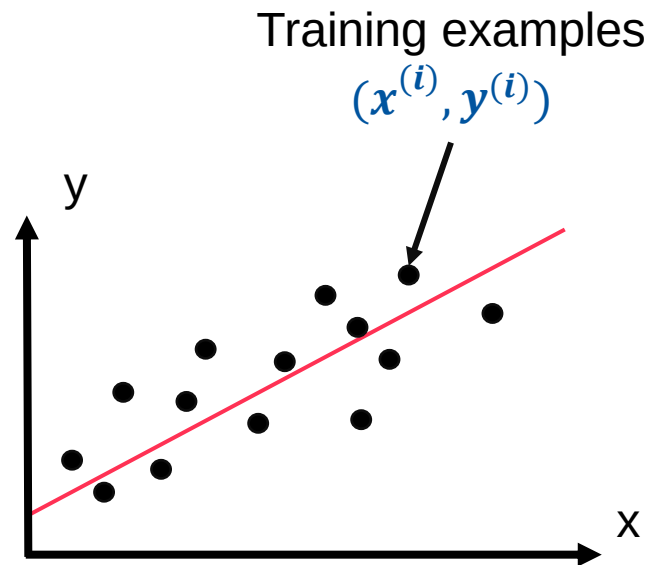


Representation of h with multiple variables

- We search for the coefficients θ to fit the regression problem

$$h(x^{(i)}) = \theta_0 + \theta_1 x_1^{(i)} + \theta_2 x_2^{(i)} + \dots + \theta_n x_n^{(i)}$$

- $(x^{(i)}, y^{(i)})$
 - used to determine the most optimum value of θ
 - known (labeled) input-output pairs
 - points of the “real” function



Representation of h with two variables

- Linear regression with two variables

$$h_{\theta}(\mathbf{x}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

$$= \sum_{k=0}^3 \theta_k x_k$$

$$= \boldsymbol{\theta}^T \mathbf{x}$$

$$\boldsymbol{\theta} = [\theta_1 \ \theta_2 \ \theta_3]$$

$$\mathbf{x} = [x_0 \ x_1 \ x_2]$$

$$x_0 = 1$$

Representation of h with n variables (multivariate)

- Multivariate linear regression with n variables

$$h_{\theta}(\mathbf{x}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$$

$$\begin{aligned} &= \sum_{k=0}^n \theta_k x_k \\ &= \boldsymbol{\theta}^T \mathbf{x} \end{aligned}$$

$$\boldsymbol{\theta} = [\theta_1 \ \theta_2 \ \dots \ \theta_n]$$

$$\mathbf{x} = [x_0 \ x_1 \ \dots \ x_n]$$

$$x_0 = 1$$

Search/optimization with cost function

- For univariate regression model:

$$\min_{\theta} J(\theta) = \arg \min_{\theta_0, \theta_1} \frac{1}{m} \sum_{i=1}^m [h_{\theta}(x^{(i)}) - y^{(i)}]^2$$

- For bivariate regression model:

$$\min_{\theta} J(\theta) = \arg \min_{\theta_0, \theta_1, \theta_2} \frac{1}{m} \sum_{i=1}^m [h_{\theta}(x^{(i)}) - y^{(i)}]^2$$

- For multivariate regression model:

$$\min_{\theta} J(\theta) = \arg \min_{\theta_0, \theta_1, \dots, \theta_n} \frac{1}{m} \sum_{i=1}^m [h_{\theta}(x^{(i)}) - y^{(i)}]^2$$

Thank you!

